The modulus function

Cambridge insert: |x| as a graph, as an equation, as an inequality — and the reason it has no derivative at the corner.

LESSON 15–16 | 2 × 40 MIN | ALGEBRA 11, §1.5 (EXTENSION) | P1 · 1.6 · P2 · 1.1–1.3

LESSON CLOCK

10' | 18' | 22' | 22' | 16' | 12'

Three ways to read |x|	10 min
Graphs	18 min
Equations	22 min
Inequalities	22 min
The corner	16 min
Homework	12 min

LEARNING OBJECTIVES

- Define |x| piecewise and sketch y = |f(x)| and y = f(|x|).
- Solve equations containing a modulus, checking for false roots.
- Solve modulus inequalities by the two standard methods.
- Explain why |x| is not differentiable at zero.

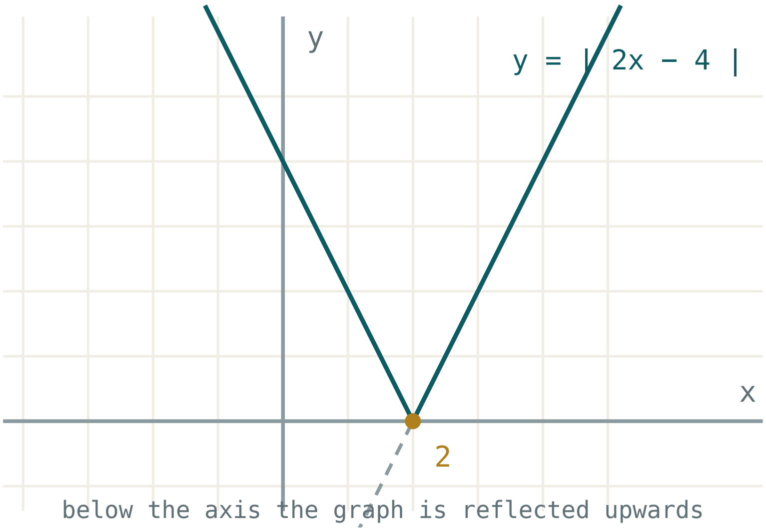
TEXTBOOK

National	pp. 75–80
Cambridge	Pure Mathematics 2 & 3, pp. 2–14
Workbook	P2 Exercise 1A–1C

01 Explanation

Three ways to read the same symbol

READING	STATEMENT
piecewise	$\lvert x\rvert = x$ if $x \geq 0$, $-x$ if $x < 0$
algebraic	$\lvert x\rvert = \sqrt{x^2}$
geometric	$\lvert x\rvert$ is the distance from x to 0; $\lvert a - b\rvert$ the distance between them



The V. Its two arms have gradient −1 and +1, and they meet at a corner.

THE GEOMETRIC READING SOLVES THE MOST

$|x - 3| < 2$ says "x is less than 2 away from 3", so $1 < x < 5$ — no algebra needed at all.

Two different graphs

GRAPH	CONSTRUCTION	EFFECT
$y = \|f(x)\|$	reflect the part below the x-axis upwards	never negative
$y = f(\|x\|)$	keep $x \geq 0$, reflect it in the y-axis	always even

For $f(x) = x^2 - 4$: $|f(x)|$ has its dip between -2 and 2 flipped into a hump; $f(|x|)$ is unchanged, because f was already even.

THEY ARE DIFFERENT GRAPHS

Take $f(x) = x - 1$. Then $|f(x)|$ is a V with vertex $(1, 0)$, while $f(|x|)$ is a V with vertex $(0, -1)$ opening upwards. Read which one is asked for.

Equations

Two methods, both valid:

1. **Cases.** $|A| = B$ means $A = B$ or $A = -B$, provided $B \geq 0$.
2. **Squaring.** $|A| = |B| \iff A^2 = B^2$ — clean when both sides carry a modulus.

$$|2x - 1| = 5 \Rightarrow 2x - 1 = 5 \text{ or } 2x - 1 = -5 \Rightarrow x = 3 \text{ or } x = -2$$

ALWAYS SUBSTITUTE BACK

$|x - 1| = 2x - 4$ gives $x = \frac{5}{3}$ and $x = 3$ by cases — but $\frac{5}{3}$ makes the right-hand side negative, and a modulus never is. Only $x = 3$ survives. Squaring is even more prone to this, because it destroys sign information.

Inequalities, and the corner

$$|A| < k \iff -k < A < k \quad |A| > k \iff A < -k \text{ or } A > k$$

The first is one double inequality; the second is genuinely two separate intervals and must be written with “or”.

Why there is no derivative at the corner. For $f(x) = |x|$ at 0:

$$\frac{|0 + \mathrm{h}| - |0|}{\mathrm{h}} = \frac{|\mathrm{h}|}{\mathrm{h}} = +1\ (h > 0),\ -1\ (h < 0)$$

The two one-sided limits are $+1$ and -1. They disagree, so the limit does not exist and no tangent can be drawn. Everywhere else $|x|$ is perfectly differentiable — the failure is at exactly one point.

02 Worked examples

EXAMPLE 1 **Solve $|3x + 2| = 8$.**

1. $3x + 2 = 8 \Rightarrow x = 2$
2. $3x + 2 = -8 \Rightarrow x = -\frac{10}{3}$
3. Right-hand side is positive, so both are valid.

ANSWER

$x = 2$ or $x = -\frac{10}{3}$

EXAMPLE 2 **Solve $|x - 4| < 3$.**

1. $-3 < x - 4 < 3$
2. $1 < x < 7$
3. Geometrically: within 3 of 4.

ANSWER

$1 < x < 7$

EXAMPLE 3 **Solve $|2x - 1| = |x + 4|$.**

1. Square both sides.
   $(2x - 1)^2 = (x + 4)^2$
2. $4x^2 - 4x + 1 = x^2 + 8x + 16$
3. $3x^2 - 12x - 15 = 0 \Rightarrow x^2 - 4x - 5 = 0$
4. $x = 5$ or $x = -1$.
   Both check.

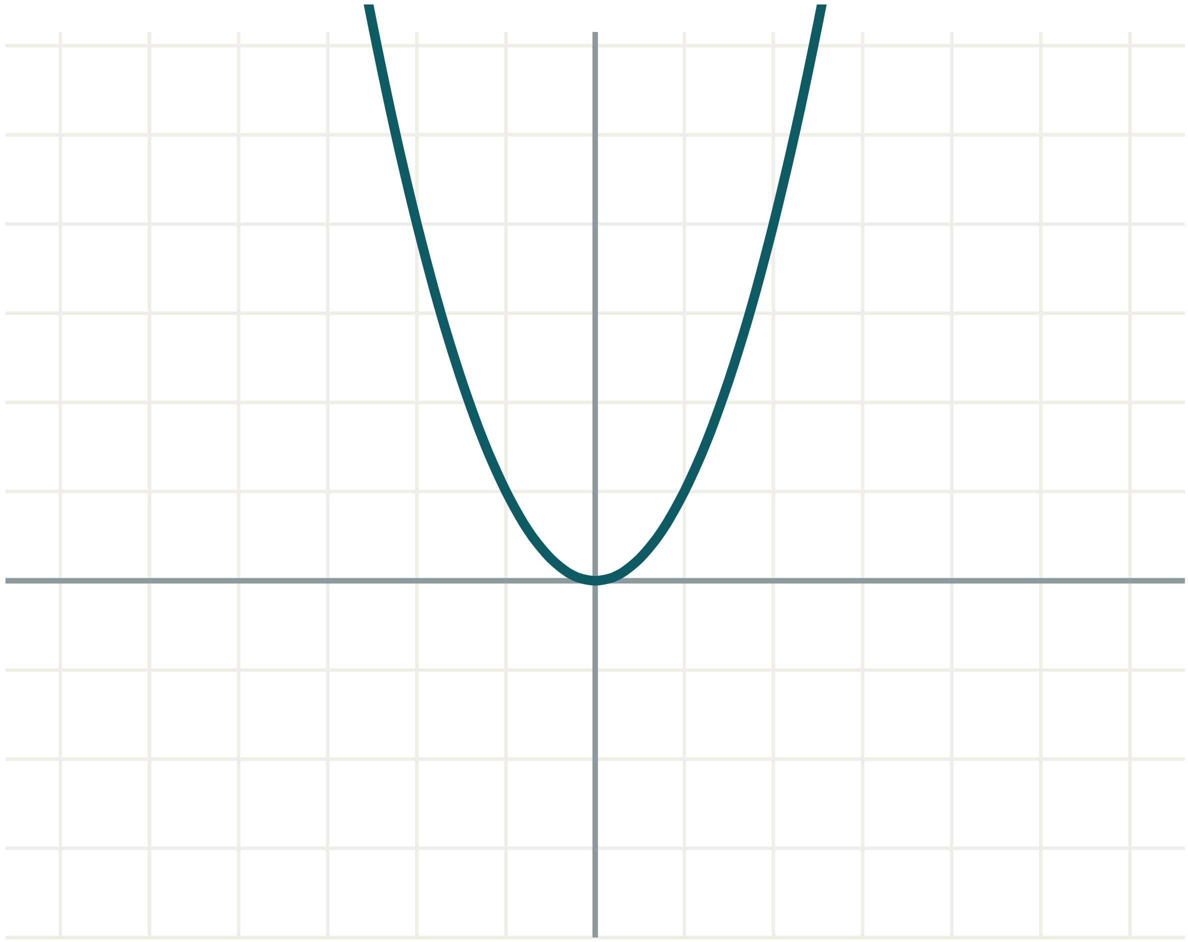
ANSWER

$x = 5$ or $x = -1$

03 Interactive model

Sketch the two sides as separate graphs and read the solutions off the intersections.

The V and its transformations



a **1** b **0** c **0** equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Modulus (absolute value)	Modul	Модуль
Piecewise definition	Bo'lakli ta'rif	Кусочное определение
Vertex of the modulus graph	Modul grafigi uchi	Вершина графика модуля
Critical value	Kritik qiymat	Критическое значение
False root	Yolg'on ildiz	Посторонний корень
Squaring both sides	Ikkala tomonni kvadratga ko'tarish	Возведение в квадрат
Distance on a number line	Sonlar o'qidagi masofa	Расстояние на числовой прямой
Corner (cusp)	Sinish nuqtasi	Угловая точка
One-sided derivative	Bir tomonlama hosila	Односторонняя производная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$|-7|$ is:

- -7
- 7
- 0
- undefined

$|x| < 4$ means:

- $x < 4$
- $-4 < x < 4$
- $x > -4$
- $x < -4$ or $x > 4$

$|x| > 4$ means:

- $-4 < x < 4$
- $x < -4$ or $x > 4$
- $x > 4$
- $x < 4$

$|x|$ at $x = 0$ has:

- derivative 0
- derivative 1
- no derivative
- derivative −1

$|x - 5| = 2$ gives:

- $x = 7$
- $x = 3$
- $x = 3$ or 7
- $x = \pm 2$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $|-9|$

   ANSWER 9

2. $|0|$

   ANSWER 0

3. Solve $|x| = 6$

   ANSWER $x = \pm 6$

4. Solve $|x - 2| = 5$

   ANSWER $x = 7, -3$

5. Solve $|x| < 3$

   ANSWER $-3 < x < 3$

6. Solve $|x| \geq 2$

   ANSWER $x \leq -2$ or $x \geq 2$

7. Vertex of $y = |x - 4|$

   ANSWER $(4, 0)$

Medium · the standard question

7 PROBLEMS

1. Solve $|2x + 1| = 7$

   ANSWER $x = 3, -4$

2. Solve $|3 - x| = 5$

   ANSWER $x = -2, 8$

3. Solve $|x + 1| < 4$

   ANSWER $-5 < x < 3$

4. Solve $|2x - 3| > 5$

   ANSWER $x < -1$ or $x > 4$

5. Solve $|x - 1| = |x + 3|$

   ANSWER $x = -1$

6. Sketch $y = |x^2 - 4|$ — where is the hump?

   ANSWER between -2 and 2

7. Vertex of $y = |2x - 6|$

   ANSWER $(3, 0)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $|x - 1| = 2x - 4$

   ANSWER $x = 3$ only

2. Solve $|2x - 1| = |x + 4|$

   ANSWER $x = 5, -1$

3. Solve $|x^2 - 4| = 3$

   ANSWER $x = \pm 1, \pm\sqrt{7}$

4. Solve $|x - 2| + |x + 1| = 5$

   ANSWER $x = -2$ or $x = 3$

5. Solve $|x| + x = 6$

   ANSWER $x = 3$

6. Solve $|x - 3| < |x|$

   ANSWER $x > 1.5$

7. Show $|x|$ has one-sided derivatives ± 1 at 0

   ANSWER $\frac{|h|}{h}$ is $+1$ then -1

07 Homework

Homework — 5 tasks

Substitute every solution back. Modulus equations manufacture false roots.

1. Solve $|4x - 5| = 11$.
2. Solve $|x + 2| \le 6$ and $|3x - 1| > 8$.
3. Solve $|x - 3| = |2x + 1|$.
4. Sketch $y = |x^2 - 9|$ and $y = |x| - 3$ on separate axes.

5. Explain, with the two one-sided quotients written out, why $|x - 5|$ has no derivative at $x = 5$.